Gauge Transformation of a Plane Electromagnetic Wave

Source: D. Tong, Electromagnetism, Problem Sheet 3, Question 5.

Consider a plane polarized electromagnetic wave described by the vector and scalar potentials,

$$\mathbf{A}(\mathbf{r},t) = \mathbf{A}_0\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)} \qquad \text{and} \qquad \phi(\mathbf{r},t) = \phi_0\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}, \tag{1.1}$$

with constant $\mathbf{A}_0$ and ϕ_0. Use Maxwell's equations to find a relationship between $\mathbf{A}_0$ and ϕ_0.

Find a gauge transformation such that the new vector potential is "transversely polarised", i.e.\
$\mathbf{A}_0 \cdot \mathbf{k} = 0$ (Coulomb gauge, $\nabla\cdot\mathbf{A} = 0$). What is the scalar potential ϕ in this gauge?

Solution. The Maxwell equations give

$$\nabla^2\phi + \frac{\partial}{\partial t}\nabla\cdot\mathbf{A} = 0, \tag{1.2}$$

and

$$\nabla\times(\nabla\times\mathbf{A}) = -\frac{1}{c^2}\left[\nabla\left(\frac{\partial\phi}{\partial t}\right) + \frac{\partial^2\mathbf{A}}{\partial t^2}\right] = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}. \tag{1.3}$$

Then, we find

$$\mathbf{k}^2\phi_0 - \omega\mathbf{k}\cdot\mathbf{A}_0 = 0. \tag{1.4}$$

If we set the Coulomb gauge, we find

$$\phi_0 = 0. \tag{1.5}$$